

SOFTWARE REQUIREMENTS SPECIFICATION

Islamic Pilgrimage Data Analysis & Insight System

Based on the Global Islamic Pilgrimage Dataset (2000–2026)

Document Type	Software Requirements Specification (SRS)
Prepared For	Data-driven software / analytics project
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Dataset Source	Kaggle — Muhammad Waqas 023
Dataset File	Islamic_Pilgrimage_2000_2026_Dataset.csv
Coverage Indicated by Dataset Name	2000–2026
Version	1.0
Status	Initial Requirements Baseline

Source reference: <https://www.kaggle.com/datasets/muhammadwaqas023/islamic-pilgrimage-dataset-2000-2026>

Important data note: This SRS is grounded in the supplied dataset identity and project context. Exact column names, data types, row count, null patterns, and value ranges should be confirmed from the CSV before implementation; they are therefore not invented in this document.

Document Control

ID	Requirement	Description
DC-01	Document owner	Hafsa Zahoor
DC-02	Version	1.0
DC-03	Status	Initial baseline
DC-04	Primary artifact	Islamic_Pilgrimage_2000_2026_Dataset.csv
DC-05	Source	Kaggle dataset supplied for this project
DC-06	Requirement basis	Dataset-driven pilgrimage analytics and reporting

Revision History

ID	Requirement	Description
1.0	Initial version	Created the baseline SRS covering scope, data requirements, functional requirements, quality requirements, interfaces, validation, and acceptance criteria.

Contents

- 1. Introduction
- 2. Overall Description
- 3. Data Requirements
- 4. Functional Requirements
- 5. Non-Functional Requirements
- 6. User Interface & Reporting Requirements
- 7. Data Quality, Validation & Error Handling
- 8. Security, Privacy & Responsible Data Use
- 9. Technical Environment & Constraints
- 10. Acceptance Criteria
- 11. Future Enhancements

Requirements notation

FR = Functional Requirement; **NFR** = Non-Functional Requirement; **DR** = Data Requirement; **UI** = User Interface Requirement; **AC** = Acceptance Criterion.

Priority levels used in this SRS are **Must**, **Should**, and **Could**. Must requirements are necessary for a usable first release.

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the requirements for an **Islamic Pilgrimage Data Analysis & Insight System**. The system will use the supplied pilgrimage dataset as its primary structured data source and provide a controlled workflow for loading, validating, preparing, analyzing, visualizing, and reporting pilgrimage-related information.

The SRS establishes a clear boundary between what the software must do and how the implementation may be designed. It is intended to guide development, testing, demonstration, and future extension.

1.2 Scope

- Import the supplied CSV dataset and verify that it is readable and structurally valid.
- Inspect dataset structure, data types, missing values, duplicates, and basic data-quality indicators.
- Prepare data for analysis without changing the meaning of source observations.
- Support year-wise and category-wise exploration of pilgrimage information where the relevant fields exist.
- Provide visual summaries and trend-oriented analytical views.
- Allow users to filter the analytical view and inspect selected records.
- Generate concise analytical outputs suitable for academic/project reporting.
- Keep the workflow reproducible so the same source data and processing rules produce consistent results.

1.3 Intended Audience

- Project developer / student implementing the analytical system.
- Instructor or evaluator reviewing correctness and usability.
- End users interested in structured exploration of pilgrimage-related trends.
- Future developers extending the system with forecasting or advanced analytics.

1.4 Definitions

ID	Requirement	Description
TERM-01	Pilgrimage record	One dataset observation representing a pilgrimage-related entry.
TERM-02	Source data	The original CSV values before transformation.
TERM-03	Processed data	Data after validated cleaning and preparation steps.
TERM-04	Analytical view	A filtered or aggregated representation used for interpretation.
TERM-05	Dashboard/report	User-facing presentation of validated metrics, tables, and visualizations.

2. Overall Description

2.1 Product Perspective

The proposed system is a dataset-driven analytical application. The CSV file acts as the authoritative input for the initial release. The application should separate data ingestion, validation, processing, analytics, visualization, and reporting so that individual components can be tested independently.

2.2 High-Level Workflow

Source CSV → Data Validation → Cleaning & Preparation → Analytical Processing → Visualization → User Filtering → Report/Export

2.3 Core Capabilities

- Reliable CSV loading.
- Schema and data-quality validation.
- Exploratory analysis.
- Time-based trend analysis where year/date fields are available.
- Category and subgroup comparisons where categorical fields exist.
- Interactive filtering.
- Summary statistics and visualizations.
- Export of filtered/processed results.

2.4 Assumptions

- The supplied CSV remains the primary source for the first release.
- The exact schema will be detected from the actual CSV rather than assumed from the filename.
- Where a field is not present, the corresponding feature will be disabled or reported as unavailable rather than fabricated.
- Source values will be preserved unless a documented cleaning rule is applied.
- The system is initially intended for analytical/educational use rather than operational pilgrimage management.

2.5 Constraints

- The system depends on the quality and completeness of the source CSV.
- Any external data source or live API is outside the baseline scope unless explicitly added later.
- Exact field-level requirements depend on the actual CSV schema.

3. Data Requirements

3.1 Source Dataset

The system shall use **Islamic_Pilgrimage_2000_2026_Dataset.csv** as its baseline input. The supplied dataset name indicates coverage from 2000 through 2026. The Kaggle page identifies the dataset as a global Islamic pilgrimage dataset. Exact field names and schema must be obtained directly from the CSV during implementation.

3.2 Schema Discovery

ID	Requirement	Description
DR-01	Column discovery	The system shall read and display the actual column names from the CSV.
DR-02	Data types	The system shall inspect inferred data types and identify fields requiring conversion.
DR-03	Row count	The system shall report the number of source records loaded.
DR-04	Null profile	The system shall report missing-value counts and percentages by column.
DR-05	Duplicate profile	The system shall report duplicate rows and support documented duplicate handling.
DR-06	Range validation	Numeric/date-like fields shall be checked for invalid or unexpected ranges.
DR-07	Source preservation	Original source data shall not be overwritten during cleaning.

3.3 Data Dictionary

A final implementation data dictionary shall be generated from the actual CSV and shall include: column name, business meaning, data type, allowed/observed range, missing-value behavior, transformation rule, and analytical use. This approach prevents incorrect assumptions about column names or meanings.

3.4 Data Processing Rules

- Trim unnecessary whitespace from textual fields.
- Normalize clearly inconsistent categorical spellings only when the correction rule is documented.
- Convert numeric fields stored as text when conversion is unambiguous.
- Convert date/year fields to an appropriate temporal representation when applicable.
- Handle missing values using field-appropriate rules and preserve a record of the action.
- Remove exact duplicate records only when justified and document the number removed.
- Never silently replace invalid values with arbitrary values.

4. Functional Requirements

4.1 Data Ingestion

ID	Requirement	Description
FR-01	Load CSV	The system must load the supplied pilgrimage CSV from a configured local path or uploaded file.
FR-02	Validate file	The system must detect missing file, unreadable file, malformed CSV, and empty dataset conditions.
FR-03	Show schema	The system must display actual columns, row count, and data types after loading.

4.2 Data Quality & Preparation

ID	Requirement	Description
FR-04	Missing values	The system must identify missing values and apply documented handling rules.
FR-05	Duplicates	The system must identify exact duplicates and report the handling outcome.
FR-06	Invalid values	The system must flag values that fail type, range, or consistency checks.
FR-07	Processed dataset	The system must produce a separate processed dataset for downstream analysis.

4.3 Exploratory Analysis

ID	Requirement	Description
FR-08	Summary statistics	The system should calculate suitable descriptive statistics for numerical fields.
FR-09	Category analysis	The system should summarize categorical fields by frequency and proportion.
FR-10	Time analysis	Where a year/date field exists, the system should support year-wise or time-wise trends.
FR-11	Comparison	The system should allow meaningful subgroup comparisons using available categorical dimensions.
FR-12	Correlation	The system should provide correlation analysis only for suitable numerical variables.

4.4 Visualization

ID	Requirement	Description
FR-13	Trend chart	Provide a line chart for valid temporal trends.
FR-14	Comparison chart	Provide bar/grouped-bar views for category comparisons.

ID	Requirement	Description
FR-15	Distribution chart	Provide histograms/box plots for suitable numerical distributions.
FR-16	Relationship chart	Provide scatter plots where two numerical variables are meaningfully related.
FR-17	Interactive filters	Allow users to filter by available year/category dimensions.

4.5 Reporting & Export

ID	Requirement	Description
FR-18	Filtered table	Display the records corresponding to the current filter selection.
FR-19	Export	Allow export of a filtered/processed dataset in CSV format.
FR-20	Summary report	Provide a concise summary of key findings and selected metrics.

5. Non-Functional Requirements

ID	Requirement	Description
NFR-01	Usability	The interface shall be simple, readable, and understandable without technical knowledge.
NFR-02	Performance	Typical filtering and visualization operations should respond promptly on a normal project dataset.
NFR-03	Reliability	Invalid files and malformed values shall produce clear messages rather than silent failures.
NFR-04	Maintainability	Data loading, cleaning, analytics, and visualization logic should remain modular.
NFR-05	Reproducibility	Processing rules and analytical calculations shall be deterministic where randomness is used.
NFR-06	Scalability	The design should allow the dataset to grow without requiring a complete rewrite.
NFR-07	Portability	The application should run in a standard Python environment with documented dependencies.
NFR-08	Accessibility	Charts and tables should use clear labels, readable text, and sufficient visual contrast.
NFR-09	Data integrity	The source dataset shall remain unchanged; transformations shall operate on a working copy.
NFR-10	Transparency	Important cleaning and transformation decisions shall be visible to the user.

5.1 Recommended Technology Stack

- **Python:** core implementation language.
- **Pandas:** CSV loading, cleaning, transformation, and tabular analysis.
- **NumPy:** numerical operations where required.
- **Matplotlib / Seaborn / Plotly:** analytical visualizations.
- **Streamlit:** optional interactive presentation layer.
- **Jupyter Notebook:** development, exploration, and academic demonstration.

6. User Interface & Reporting Requirements

6.1 Main View

- Clear project title and short description.
- Dataset status showing loaded/not loaded state.
- High-level metrics such as records, fields, missing values, and duplicates.
- Interactive filters based only on fields that actually exist.
- Analytical charts and supporting tables.

6.2 Suggested Information Architecture

ID	Requirement	Description
UI-01	Overview	Dataset summary, coverage, record count, and key indicators.
UI-02	Data Quality	Missing values, duplicates, type/range checks, and cleaning summary.
UI-03	Trends	Temporal analysis when date/year fields are available.
UI-04	Comparisons	Category/subgroup comparisons using available fields.
UI-05	Relationships	Numerical relationships and distributions.
UI-06	Records	Searchable/filterable data table.
UI-07	Export	Download filtered or processed data.

6.3 Reporting Principles

- Every chart shall have a meaningful title and axis labels.
- Units shall be shown where applicable.
- Aggregations shall state the measure being summarized.
- Charts shall not imply relationships that the data cannot support.
- Missing or unavailable dimensions shall be explicitly indicated.

7. Data Quality, Validation & Error Handling

7.1 Validation Checks

ID	Requirement	Description
VAL-01	File validation	Confirm that the input exists and can be parsed as a CSV.
VAL-02	Schema validation	Confirm that required analytical fields exist before enabling dependent features.
VAL-03	Type validation	Detect numeric, categorical, and temporal fields that contain incompatible values.
VAL-04	Missingness	Report null count and percentage for each column.
VAL-05	Duplicates	Report exact duplicate rows and identify the final handling action.
VAL-06	Range checks	Flag impossible or suspicious numeric/date values where domain rules are available.

7.2 Error Handling

- If the CSV cannot be opened, show a clear file error and stop downstream processing.
- If a requested filter field does not exist, disable that filter rather than raising an unhandled exception.
- If a chart cannot be produced because the required data type is unavailable, show an explanatory message.
- If a calculation encounters missing/invalid values, report the affected records or use a documented exclusion rule.
- All user-facing errors should be concise and actionable.

7.3 Auditability

The system should retain a compact processing summary: source row count, duplicate count, missing-value actions, rows removed if any, and final processed row count. This creates a reproducible audit trail for academic evaluation.

8. Security, Privacy & Responsible Data Use

The baseline project is an analytical dataset application. It should nevertheless follow responsible data practices.

- Do not collect personal information unless it is explicitly required by a future scope.
- Do not expose local file paths or environment secrets in the user interface.
- Do not embed credentials or API keys directly in source code.
- If external services are introduced later, store credentials in environment variables or a secure secret store.
- Keep exported data limited to information appropriate for the intended audience.
- Present analytical results as observations from the dataset rather than unsupported claims about individual pilgrims or communities.

8.1 Ethical Analytics

Pilgrimage data can involve sensitive religious and demographic context. The system should therefore avoid discriminatory inference, unnecessary profiling, or conclusions that go beyond the evidence contained in the dataset. Aggregated analysis is preferred where possible.

9. Technical Environment & Constraints

ID	Requirement	Description
TECH-01	Runtime	Python 3.x environment.
TECH-02	Input	CSV dataset: Islamic_Pilgrimage_2000_2026_Dataset.csv.
TECH-03	Data layer	Pandas/NumPy recommended.
TECH-04	Visualization	Plotly, Matplotlib, or Seaborn depending on implementation.
TECH-05	Interface	Streamlit may be used for an interactive application.
TECH-06	Development	Jupyter Notebook may be used for exploratory and academic work.
TECH-07	Versioning	Project code and requirements should be maintained under version control.

9.1 Suggested Project Structure

```
islamic-pilgrimage-analysis/
├── data/
│   └── Islamic_Pilgrimage_2000_2026_Dataset.csv
├── notebooks/
│   └── exploration.ipynb
├── src/
│   ├── data_loader.py
│   ├── preprocessing.py
│   ├── analysis.py
│   └── visualization.py
├── app.py
├── requirements.txt
└── README.md
```

9.2 Deployment Readiness

- Dependencies should be pinned or version-bounded in requirements.txt.
- The application should run from a clean environment.
- The README should document installation, dataset placement, execution, and known limitations.
- No hard-coded machine-specific paths should be required for deployment.

10. Acceptance Criteria

The first release will be considered acceptable when the following conditions are met:

ID	Requirement	Description
AC-01	Dataset loading	The supplied CSV loads successfully and the system reports its actual schema and record count.
AC-02	Data quality	Missing values and duplicates are detected and their handling is reported.
AC-03	Processing	A processed working dataset is generated without overwriting the source.
AC-04	Analysis	At least one valid numerical summary and one valid categorical/time analysis are available when supported by the schema.
AC-05	Visualization	Relevant charts render with correct labels and no fabricated dimensions.
AC-06	Filtering	Available analytical dimensions can be filtered without unhandled errors.
AC-07	Export	A filtered/processed table can be exported successfully.
AC-08	Reproducibility	Running the same source data and processing configuration produces consistent results.
AC-09	Usability	A new user can understand the main dashboard/report without reading source code.
AC-10	Documentation	The project contains clear setup, data-source, processing, and usage documentation.

11. Future Enhancements

- Automated ingestion of updated pilgrimage datasets.
- Advanced time-series forecasting for supported numerical measures.
- Interactive geographic analysis if location-level fields are available.
- Role-based access for administrative deployments.
- Scheduled report generation.
- Comparative analysis across pilgrimage years and available demographic/category dimensions.
- Anomaly detection for unusual changes in yearly or category-level measures.
- Multilingual interface support.

11.1 Requirement Traceability Principle

Each future enhancement should be converted into an explicit requirement with a unique ID, acceptance criterion, and test case before implementation. This keeps the project maintainable as the analytical scope grows.

End of Software Requirements Specification